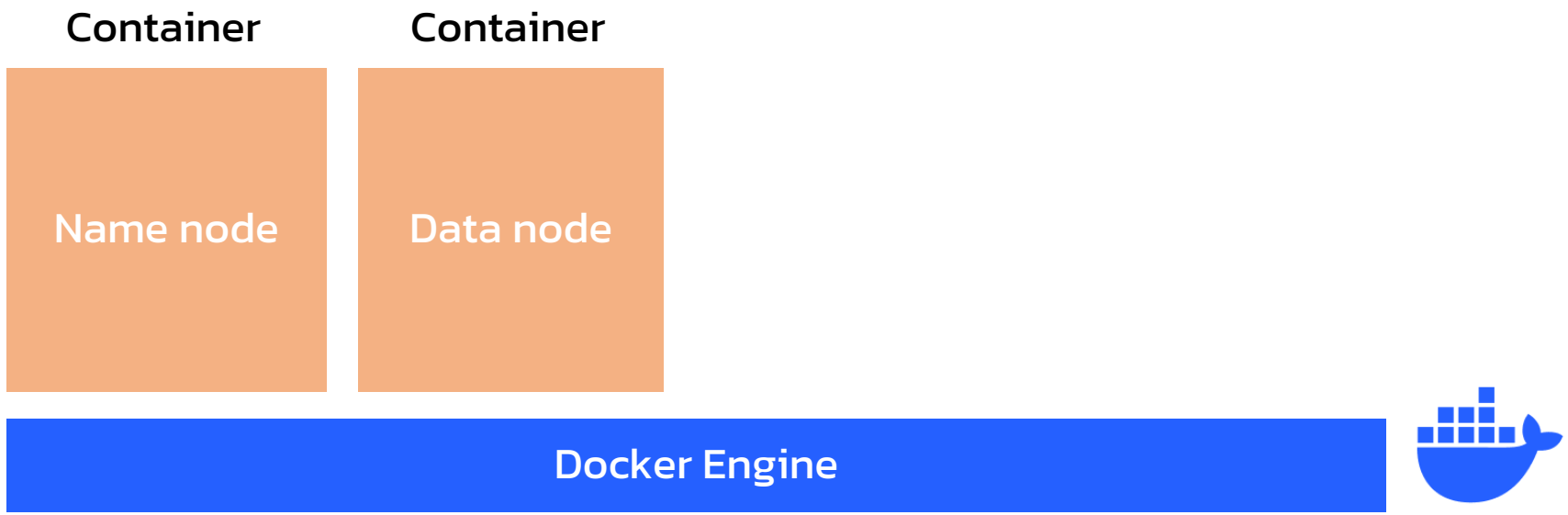
# HDFS Installation & Basic Operation

# Topic

- Build the Docker Image
- Docker container basic
- Build the Docker Compose file
- Start Docker Compose
- HDFS Basic Operation

# Building the Docker Image

# The Architecture Concept



# Basic Docker Build

## Basic Docker Build image

```
docker build -t <tagname : tagversion> .

#Ex
docker build -t my-first-image : 0.1 .
```

Build, tag, and publish an image | Docker Docs

Use lab0 file

# Prepare docker file for Name node & Data node

## Name node docker file

dockerfile-namenode

```
FROM bde2020/hadoop-namenode:2.0.0-hadoop3.2.1-java8

RUN echo "deb http://archive.debian.org/debian/ stretch main" > /etc/apt/sources.list && \
    echo "deb http://archive.debian.org/debian-security/ stretch/updates main" >> /etc/apt/sources.list && \
    echo "Acquire::Check-Valid-Until \"false\";" > /etc/apt/apt.conf.d/99no-check-valid-until && \
    apt-get update && apt-get install -y python3
```

## Data node docker file

dockerfile-datanode

```
FROM bde2020/hadoop-datanode:2.0.0-hadoop3.2.1-java8

RUN echo "deb http://archive.debian.org/debian/ stretch main" > /etc/apt/sources.list && \
    echo "deb http://archive.debian.org/debian-security/ stretch/updates main" >> /etc/apt/sources.list && \
    echo "Acquire::Check-Valid-Until \"false\";" > /etc/apt/apt.conf.d/99no-check-valid-until && \
    apt-get update && apt-get install -y python3
```

Use lab0 file

# Build The Name node & Data note Docker Image

## Build Name node image

```
docker build -f dockerfile-namenode -t hadoop-namenode-python:0.1 .
```

## Build Data node image

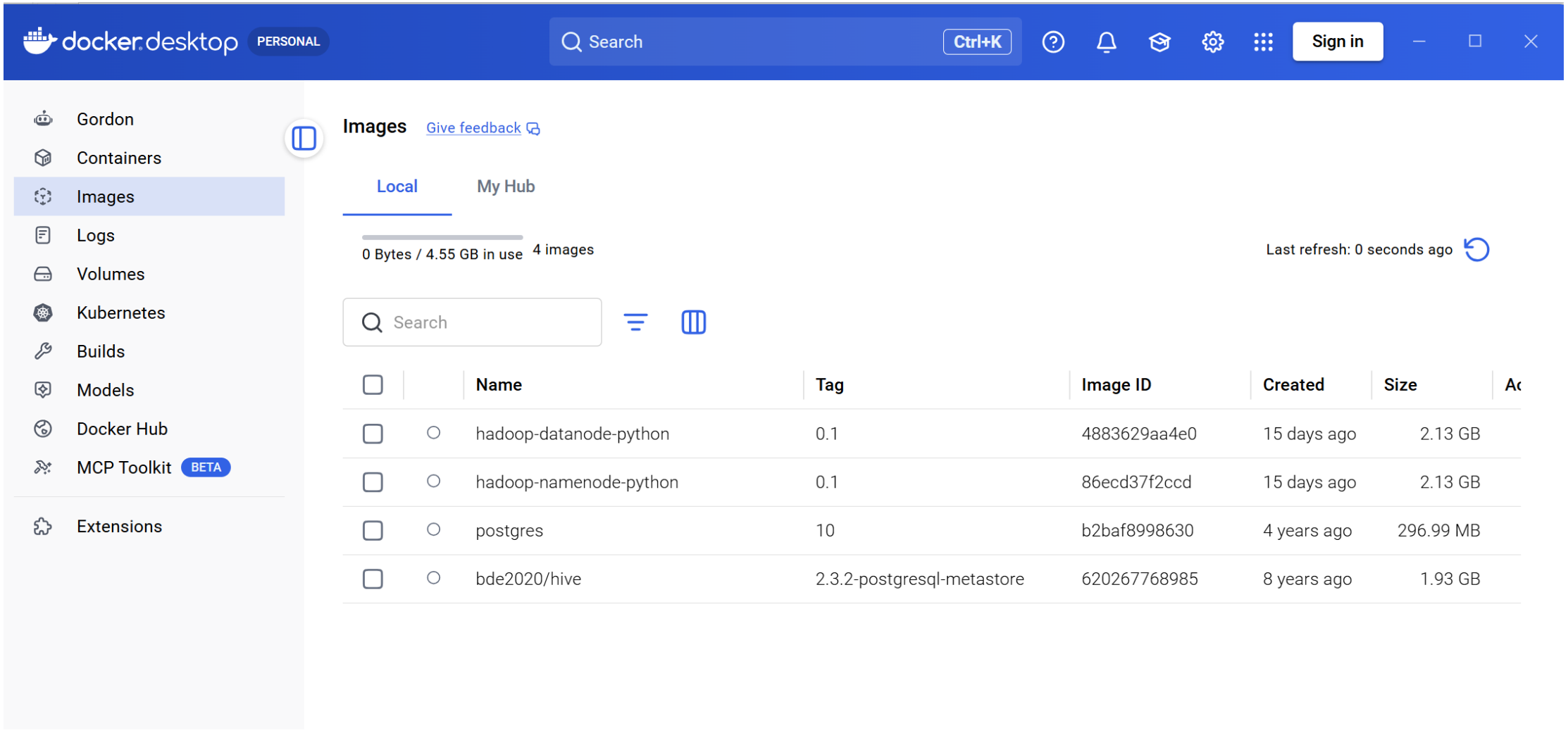
```
docker build -f dockerfile-datanode -t hadoop-datanode-python:0.1 .
```

# Image output

```
C:\Users\TU>docker image ls

IMAGE                                          ID             DISK USAGE   CONTENT SIZE   EXTRA
bde2020/hive:2.3.2-postgresql-metastore        620267768985       1.93GB          684MB
hadoop-datanode-python:0.1                     4883629aa4e0       2.13GB          589MB
hadoop-namenode-python:0.1                     86ecd37f2ccd       2.13GB          589MB
postgres:10                                    b2baf8998630        297MB         74.1MB
```

# Image output (#2)

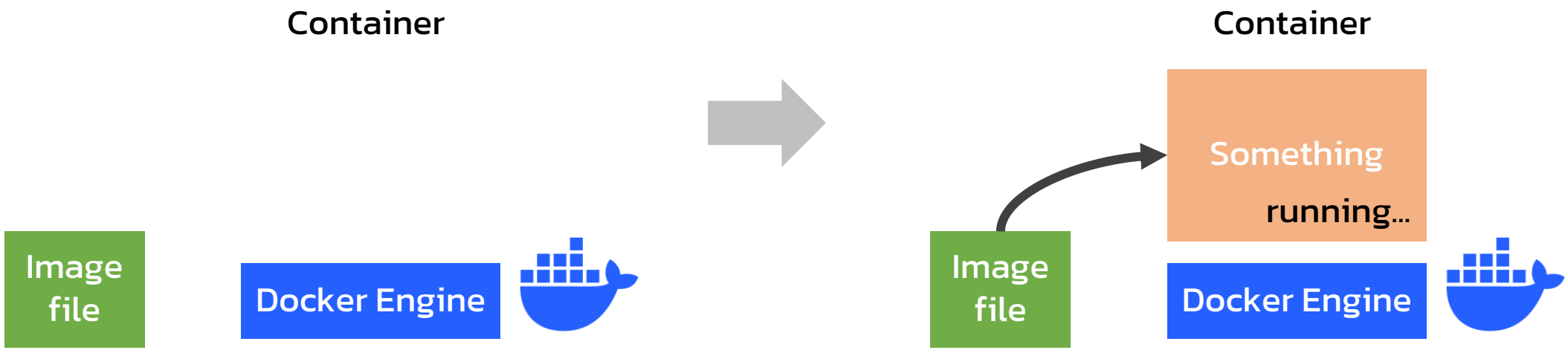


Use lab0 file

# Docker container basic

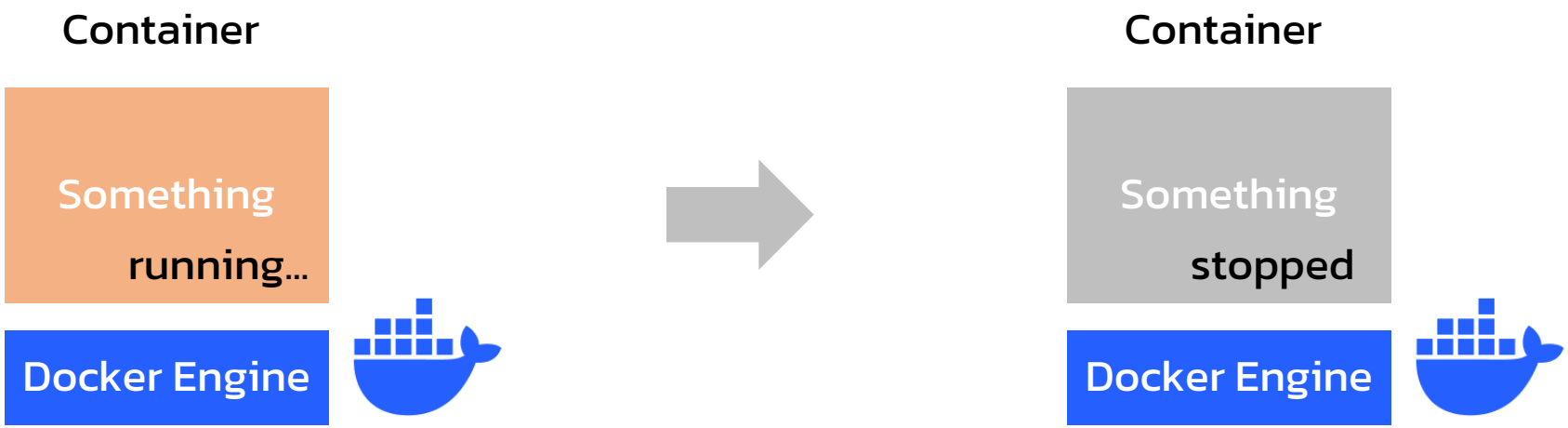
Docker run <option> image_name

option [-it , --name]

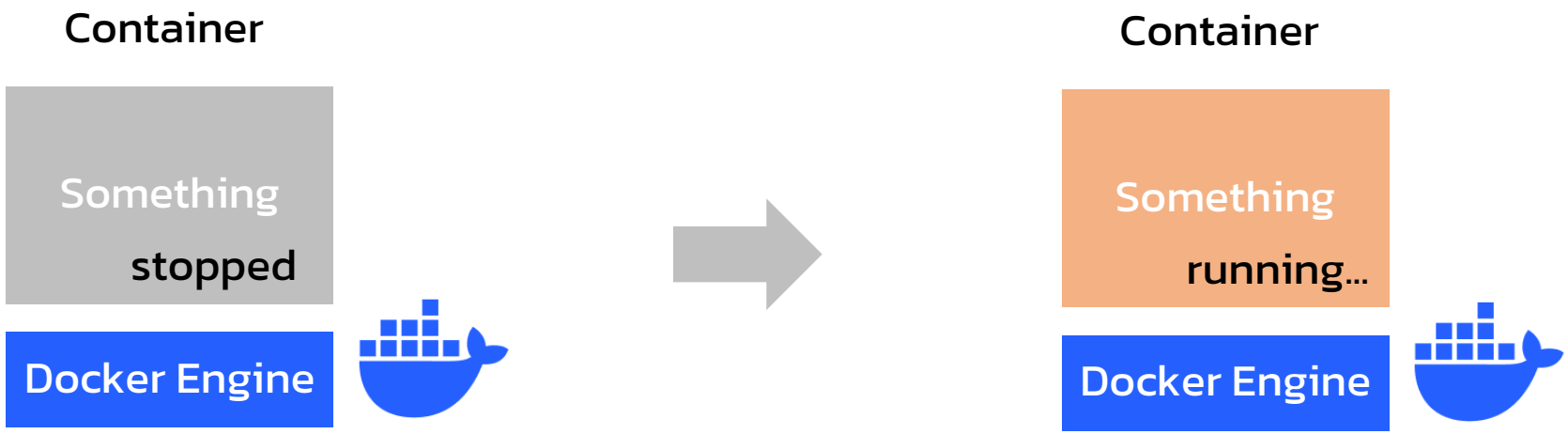



*สร้าง Container ใหม่จาก Image แล้วสั่งเปิดทันที

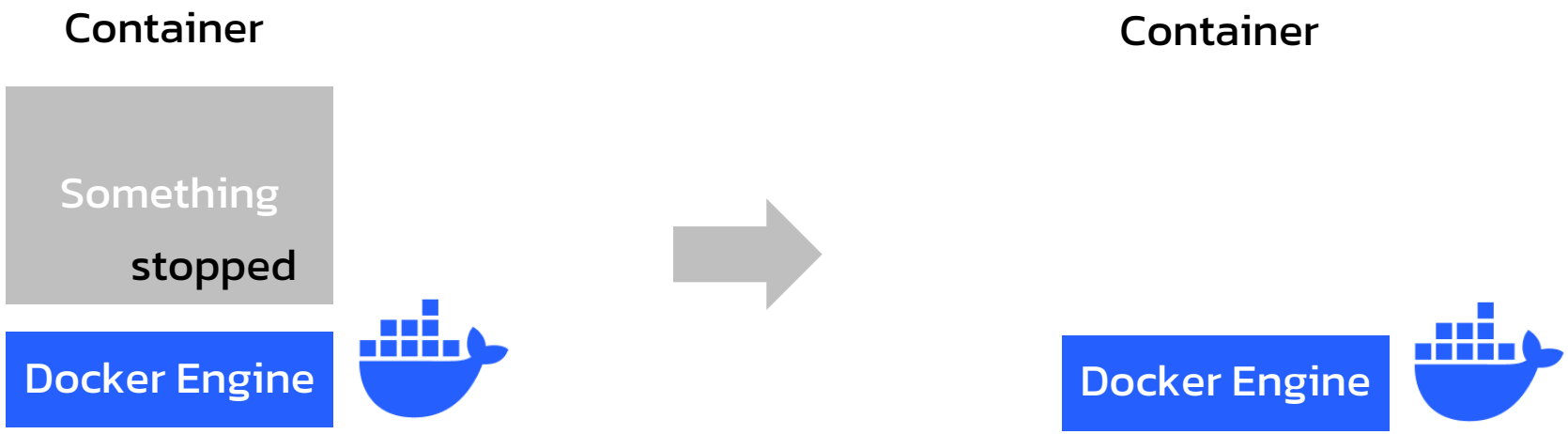
# Docker stop <container_id, container_name>



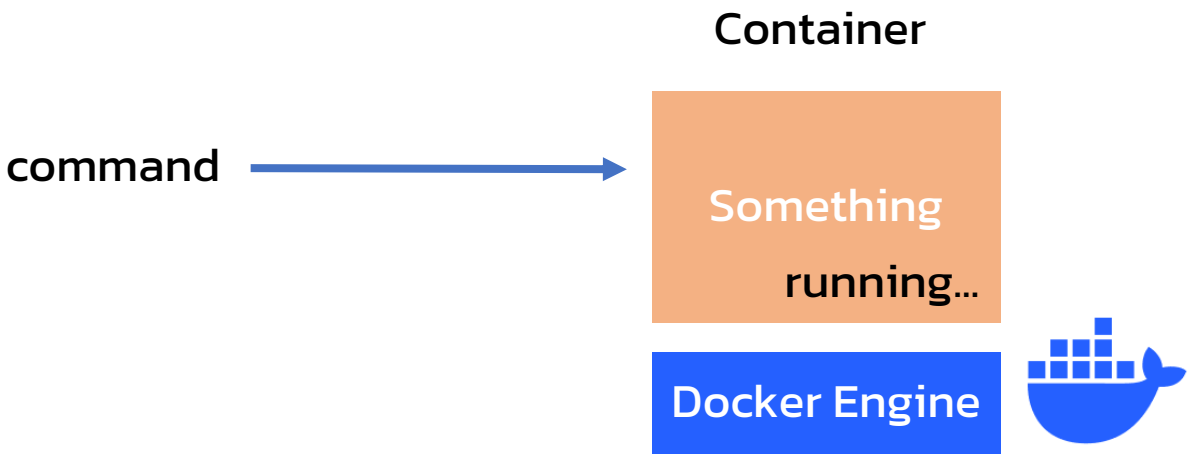
# Docker start <container_id, container_name>



# Docker rm <container_id, container_name>



# Docker exec <option> <container_id, container_name> <command>

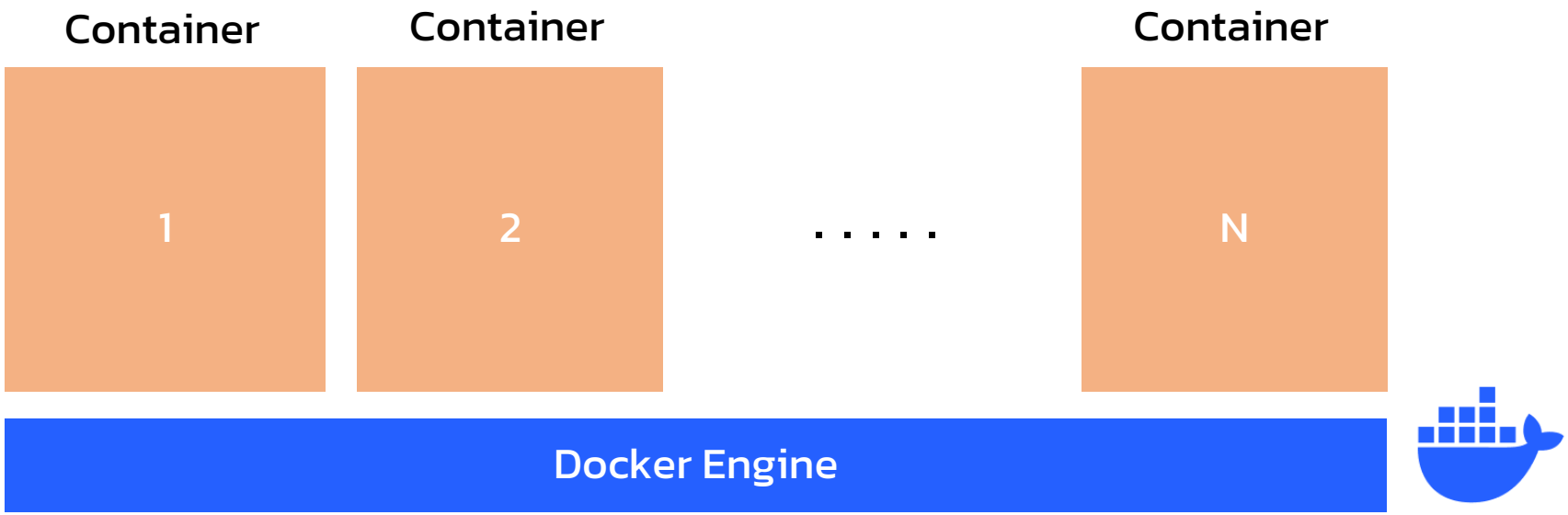


Use lab1 file

# Docker compose file

# Why do we use docker compose ?

Some time our system may not use single container, but we use many ....



# Hadoop Lap Docker compose file argument

01-docker-hdfs-compose.yml

```yaml
name: "my-hadoop"

version: "3"

services:
  namenode:
    image: hadoop-namenode-python:0.1
    container_name: namenode
    ports:
      - "9870:9870"
      - "9000:9000"
    volumes:
      - hadoop_namenode:/hadoop/dfs/name
    environment:
      - CLUSTER_NAME=test
    env_file:
      - ./hadoop.env

  datanode:
    image: hadoop-datanode-python:0.1
    container_name: datanode
    volumes:
      - hadoop_datanode:/hadoop/dfs/data
    env_file:
      - ./hadoop.env

volumes:
  hadoop_namenode:
  hadoop_datanode:
```

compose name

compose version

services

service

service

# Hadoop Lap Docker compose file argument

01-docker-hdfs-compose.yml

```yaml
name: "my-hadoop"

version: "3"

services:
  namenode:
    image: hadoop-namenode-python:0.1
    container_name: namenode
    ports:
      - "9870:9870"
      - "9000:9000"
    volumes:
      - hadoop_namenode:/hadoop/dfs/name
    environment:
      - CLUSTER_NAME=test
    env_file:
      - ./hadoop.env

  datanode:
    image: hadoop-datanode-python:0.1
    container_name: datanode
    volumes:
      - hadoop_datanode:/hadoop/dfs/data
    env_file:
      - ./hadoop.env

volumes:
  hadoop_namenode:
  hadoop_datanode:
```

Docker image

Container name

การ Binding Port <host> : <container>

Environment setup

Environment setup from file

Volumes <volumes > : <container path>

Volumes

test lab1 file

# Start and Stop Docker compose

## Start Docker Compose

```
docker compose [-f compose_file] up
```

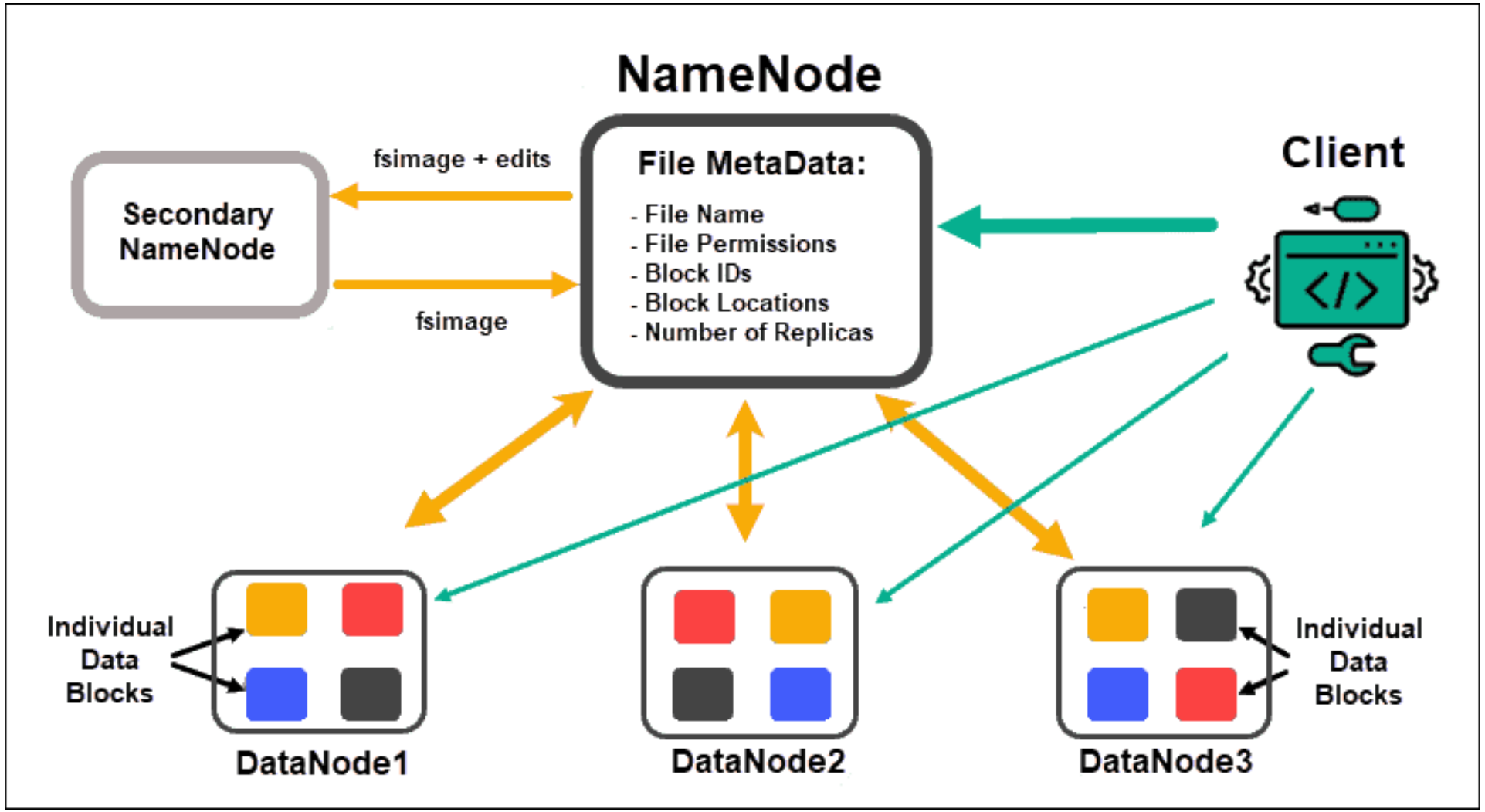
## Stop Docker Compose

```
docker compose [-f compose_file] donw
```

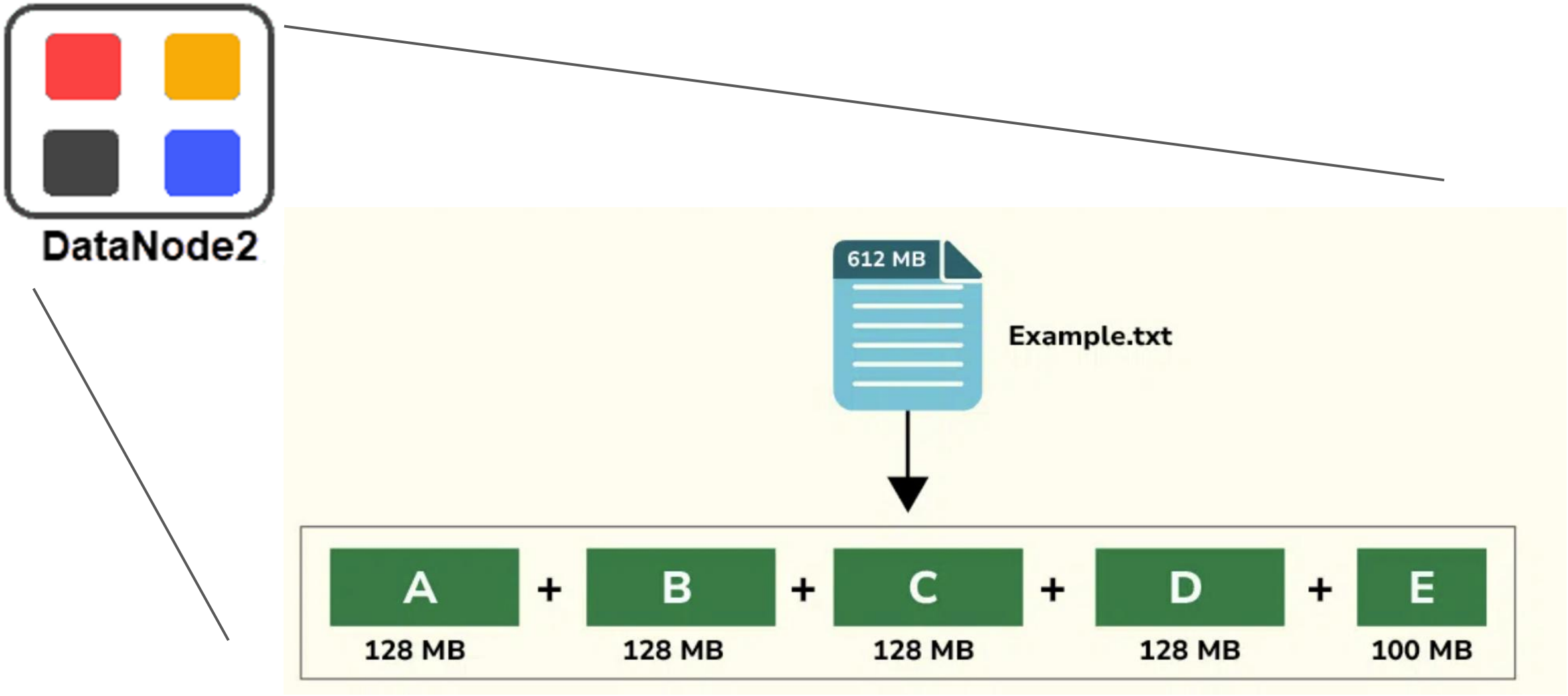
Use lab1 file

# HDFS Basic Operation

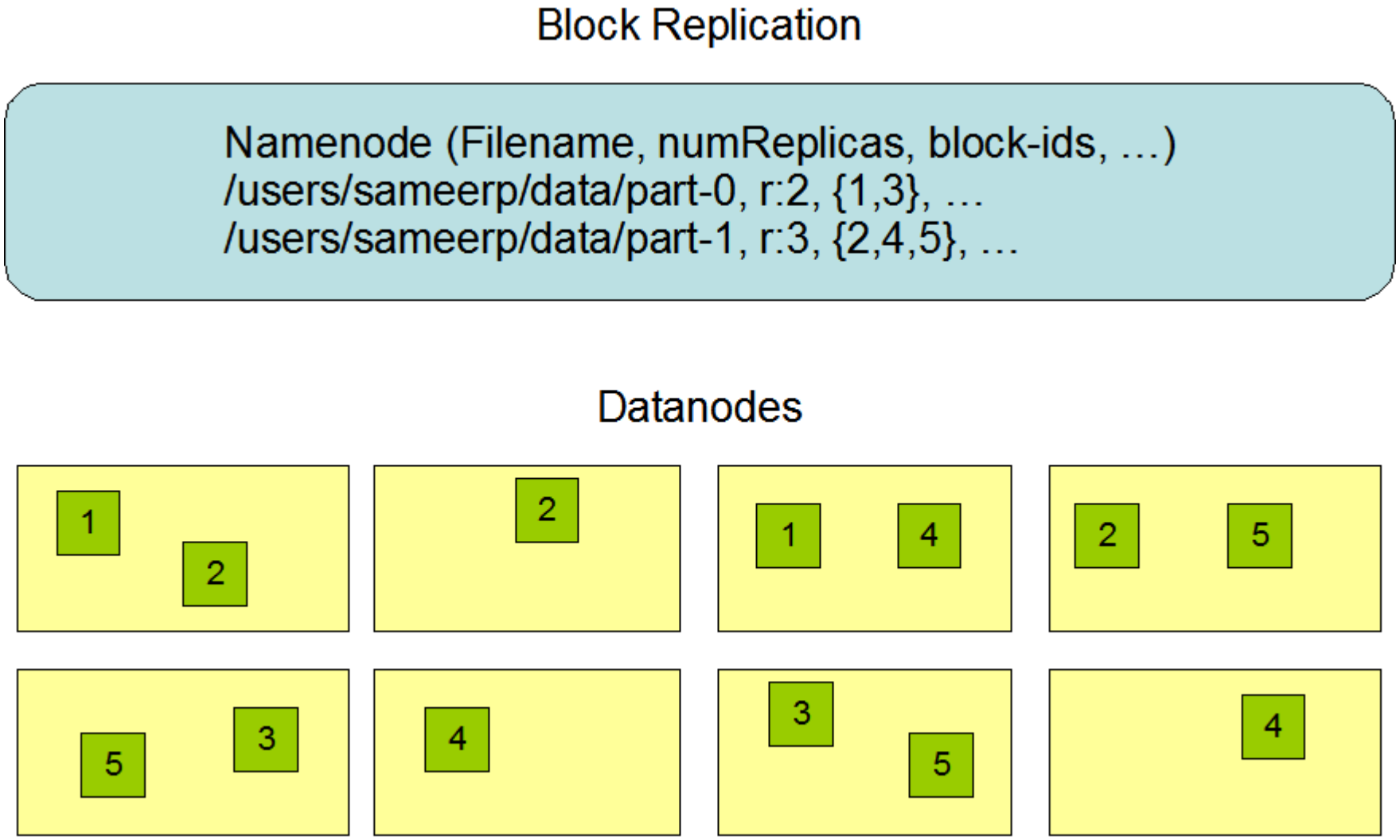
# hdfs (Hadoop distribution file system)



# hdfs data block



# hdfs data block replication



# HDFS Basic Operation

- hdfs dfs –ls <directory> : file or directory list
- hdfs dfs –mkdir <directory_name> : create directory on hdfs
- hdfs dfs –put <file> : upload file to hdfs
- hdfs dfs –cat <file> : read text file on hdfs
- hdfs dfs –get <file> : download file from hdfs
- hdfs dfs –cp <file_on_hdfs> <new_file_on_hdfs> : copy 1 file to new 1 file
- hdfs dfs –cp –r <dir_on_hdfs> <new_dir_on_hdfs> : copy 1 dir & file to new dir
- hdfs dfs –rm <file> : remove file on hdfs
- hdfs dfs –rm –r <directory> : remove directory on hdfs

# HDFS Basic Operation (cont)

- hdfs dfs –df [-h] : view disk free of hdfs
- hdfs dfs –du [-h] <file or directory> : view file or directory disk used on hdfs

End